Enabling genome-wide genealogies in complex plant genomes

Abstract
Plant genomes exhibit remarkable presence-absence variation, with the majority of annotated genes absent in at least one individual in any reasonable sample. Compounding this structural variation are the activities of transposable elements, sequences that can actively move within genomes and make up as much as 90% of many plant genomes. Recent cutting-edge efforts to study the evolutionary history of genomes have developed approaches to estimate the ancestral recombination graph (ARG), a data structure that describes how genetic material is passed from ancestors to descendants. Accurate inference of the ARG provides the opportunity to know nearly the complete evolutionary history of a sample, from selection and drift to recombination and mutation. Unfortunately, ARG inference approaches assume the sequence studied is present in all individuals and are thus unable to be used for the majority of plant genes or genomic space. The current proposal will develop methods to adapt ARG inference to regions of the genome with presence-absence variation, thus enabling a more detailed evolutionary study of the majority of plant genomes.

Background
For the past six decades, population genetics has focused on leveraging patterns of genetic variation to reconstruct past evolutionary events. Yet evolutionary inference remains challenging because the state space of possible histories is vast and methods typically rely on summary statistics rather than complete genomic data. Recent breakthroughs, however, now allow us to infer the full genealogical history of samples as a data structure called the ancestral recombination graph (ARG)[10–15].

In principle, the complete historical information contained in a set of DNA samples can be encoded as an ARG[16]. In the absence of recombination, a genealogy is the tree that describes relationships between samples. When recombination occurs, however, the genealogy is not a tree but rather a graph that encodes the transmission of genetic material from ancestors to extant individuals across the genome. Mutations can occur along the branches in this graph, yielding the observable patterns of variation. Though genomic data contains insufficient information to infer the true ARG, modern methods have successfully used simplifying assumptions to accurately infer either the posterior distribution of ARGs or a single, best guess ARG[10–12,15].

ARGs promise to change evolutionary inference in fundamental ways. Traditional inference approaches rely on summaries of the genotype matrix, such as the site frequency spectrum (SFS), nucleotide diversity, or fixation indices (*F*-statistics). But it can be challenging to distinguish between alternative scenarios and account for interactions among processes using summaries[17,18,8]. For example, diversity is impacted similarly by both positive and negative selection[19–21]. However, these processes have radically different impacts on genealogies: while positive selection leads to a burst of quick coalescences as a beneficial allele increases in frequency[22], negative selection does not[23]. Some methods circumvent the limitations of single summary statistics by combining them[24], but we lack a theoretical understanding of how processes impact statistics at different genomic and temporal scales[25,26]. ARGs, on the other hand, contain all the knowable evolutionary information from the genetic data and organize this information in such a way that reflects both the spatial and temporal scales of evolutionary events. Another benefit of using ARGs is that any statistic calculated based on allele frequencies has an equivalent and less noisy “branch” statistic (i.e., computed on the branches of the ARG)[27,28].

Though the revolution in ARG inference is in its infancy, inferred ARGs have already been successfully used to tackle various evolutionary questions. Estimating the demographic history of populations (including splits and migration) typically requires a pre-specified demographic model, limiting its usefulness in species lacking detailed historical records. Researchers developed a model-free method to infer the population size history using ARGs, shedding light on the effects of agriculture on a pollinator species history[11,29]. Another question of great interest in evolutionary biology is understanding the role of standing variation in adaptation[30]. Researchers were able to demonstrate that the mutation in a domestication locus in maize, *tga1*, was actually already present in the wild ancestors of maize, highlighting the importance of standing variation in driving adaptation during domestication. Lastly, a major leap ARGs have enabled is the inference of geographic locations of genetic ancestors over time. Most population genetic models discretize species into populations, but many species experience continuous spatial structure rather than discrete boundaries. Using ARGs, researchers were able to trace the dispersal history of humans and inferred changes in dispersal rates for *Arabidopsis thaliana* across its range[31–33].

Despite its promises, the full potential of ARGs in evolutionary inference remains unrealized, specifically because existing inference methods assume structurally stable genomes. These methods were developed primarily for human genomes, which are relatively well-behaved, with low genetic diversity and only moderate repetitive content[34,35]. However, many plant species possess much more complex genomes (**Fig. 1**), with active transposable elements and extensive presence-absence variation (PAV; genomic sequences that are absent in one or more individuals)[36–38]. In a recent study that assembled 26 maize genomes using long-read sequencing, only 15% of the genome and ~25% of genes are present in all individuals[36]. As sample size increases, the portion of the genome without PAV (or core genome) decreases even further. Because current ARG inference methods require no missing data, they cannot recover the genealogical relationships in regions affected by PAV. This precludes the study of evolutionary processes shaping most of the genome, including duplicated genes, transposable elements, and noncoding regulatory sequences. Moreover, because regions of the genome without PAV will be enriched for conserved sequences, evolutionary inferences that rely on neutral variation will often be misled[26,39]. It is thus essential to accommodate structural variation into ARG inference for it to be useful in empirical studies of plant evolution.

Here, we propose to develop a new method that enables ARG inference in complex genomes and apply it to reconstruct the genomic history of maize. Maize is an ideal model system for this work: while it exhibits genomic features that challenge current methods (such as extensive PAV), hundreds of high-quality long-read assemblies necessary to test our new approach are available (with more efforts underway)[36,40]. We will infer a whole-genome genealogy for maize now incorporating thousands of gene PAV and tens of thousands of transposable element PAV, which were previously refractory to genealogy estimation. This ARG will allow us to build on genealogy-based approaches to study selection on TE evolution[41], and inform models of linked selection[42,43].

Aim 1: Integrating structural variants into ARG inference

Rationale: Current ARG inference methods can only analyze single-nucleotide polymorphism (SNP) data from regions of the genome in which a sequence is present in all individuals in the sample. Though this is reasonable for humans, where little of the genome is impacted by PAV[44,45], many plant genomes present extensive structural variation[37,38,46]. By focusing on these regions, we are introducing biases towards structurally stable regions of plant genomes, which are enriched for sequences under increased selective constraint. PAVs have a disproportionately large impact on phenotype across many species[47,48], so they play important roles in adaptation and speciation[49]. Thus, much of the important evolutionary history and functionally relevant variation in plant genomes is missed by an inferred ARG that encompasses only SNPs in regions without PAV.

Recent advances in long-read sequencing have enabled high-quality pan-genome assemblies, revealing most of the existing structural variation[37,38,45,46]. These new assemblies provide a timely opportunity to modify ARG inference to account for PAV. In this aim, we will develop a heuristic method to accommodate PAVs in ARG reconstruction, enabling inference beyond the conserved core parts of a genome.

Approach: ARG reconstruction is computationally intensive, partly because of the large state space of possible histories. With structural variants, algorithmic complexity increases even further. Attempts at resolving this by incorporating a genome graph into ARG representation have stalled due to computational constraints.

Here, we plan to take an alternate approach that approximates the evolution of structural variants. We will construct a linear version of the pangenome, which includes the common structural variants (**Fig. 2A**). Using this linearized reference, we will obtain SNP and PAV genotypes for all individuals. Because current ARG tools do not incorporate PAVs, we will instead add temporary marker mutations along the PAV to track it. This will ensure that the ARG inference maintains a clade of samples with the PAV throughout its span. The marker mutations will be used for genealogical reconstruction together with true SNPs. Then, the inferred genealogies will be re-dated[50] without the marker mutations to ensure unbiased branch lengths. These genealogies will be post-processed to remove edges spanning a PAV in haplotypes that do not carry the PAV allele using custom code. We will make available an open-source pipeline with proper documentation that, given a pangenome and population parameters (e.g., mutation rate), will infer an ARG that includes regions with PAV.

The heuristic proposed above is approximate because of misspecification in the recombination process, but the recombination process modeled in ARG inference is approximate itself, and ARG inference tools are able to recover reductions in recombination rates driven by other structural variants such as inversions[51]. Thus, this misspecification of the recombination process will not be strong enough to meaningfully bias the inferred

genealogies. Nevertheless, we will test our approach using realistic forward-in-time simulations of genomes with structural variants in SLiM (**Fig. 2B**). These will reveal any potential biases and allow us to correct them, for instance by adjusting the number of mutations that tag PAVs. We also have plans to modify current threading-based ARG inference algorithms[15] to include PAVs. For insertions, this would entail (i) disabling recombination events involving lineages that don't carry an insertion over the span, (ii) skipping branches that do not subtend one of the insertion-carrying haplotypes when computing the mutational likelihood, and (iii) inserting a node as the root at the trees spanning the insertion. Although we know how to include at least simple PAVs into the model, it is not clear how these changes would affect software efficiency.

Aim 2: Inferring selection on non-core regions of the maize genome using whole-genome genealogies
Rationale: There is growing interest in using ARGs for evolutionary inference[52–54,33,55,56]. However, plant genomes contain substantial presence-absence variation (PAV), which limits the extent of the genome that is amenable to ARG inference to less than 30% across several species (**Fig. 1B**). Here, we will apply our new approach to explore the evolutionary processes that shape the non-core portion of the maize genome. While the maize genome is highly complex (**Fig. 1C**), it is not unusual among plants (**Fig. 1A**), making it a good model for understanding the evolution of PAV regions in plant genomes.

The observation of extensive PAV in plant genomes is puzzling. On the one hand, genes with PAV are enriched among some functional categories, such as biotic defense[57–59]. However, many of the PAVs are young and present signatures of relaxed constraint[58], suggesting that some PAVs are just transient polymorphisms that arise frequently in regions of high genomic stability (e.g., affected by more frequent transposable element insertions).

Our approach from Aim 1 will allow, for the first time, accurate genealogy inference in regions with insertions, deletions and duplications (**Fig. 2B**). We will use these genealogies to characterize the roles of purifying and balancing selection in these regions. We hypothesize that most of these regions will be under less selective constraint, but we expect to find some PAVs that have been segregating for a long time, which would point to the role of balancing selection. Indeed, a few PAVs caused by transposable elements (TEs) are under balancing selection in different species[60–62], similar to those seen in *R*-genes in *Arabidopsis*[57].
Approach: We will reconstruct whole-genome genealogies for hundreds of maize assemblies that are currently or will soon be publicly available. This will entail feeding the pangenome graph into our pipeline developed in Aim 1. We will validate the inferred ARG with realistic simulations of maize, including maize-specific recombination maps and demography. Then, we will assess the amount of site and branch diversity[27] within PAV regions. Our ARGs will enable us to evaluate the relationship between age and diversity: this is expected to be positive in neutral regions of the genome, whereas increased constraint will lead to low diversity regardless of age, and balancing selection will lead to elevated diversity and the maintenance of extremely old polymorphisms[50]. By incorporating the extensive functional genomic data available for maize[36,40,63], we can relate these population genomic signatures to features such as chromatin accessibility, gene expression, and phenotype. We will also characterize selection acting on TEs specifically and compare TEs based on transposition mechanism, family, distance to genes, etc. Our current knowledge of selection on TEs is particularly limited by the lack of PAVs on ARGs[41], given few TEs are part of the core maize genome (**Fig. 1C**). With our approach, we will move from being able to analyze 10%[41] to more than 70% of the TEs in maize, allowing for a less biased view of the role TEs play in plant genomes.

Future career goals
My long-term goal is to lead a research group that develops and applies population genomics methods to diverse biological systems. This fellowship will provide me with training in plant evolutionary genetics, particularly in the challenges of applying population genetic tools like ARG inference in complex plant genomes. My background in developing realistic simulation and ARG manipulation tools and applying them to understand empirical data will be directly applicable in building a new ARG inference approach for plant genomes. The proposed research, combined with my prior experiences, will put me at the forefront of this nascent but promising field of research of ARG-based evolutionary inference and prepare me for a faculty position in evolutionary genomics.

Figures

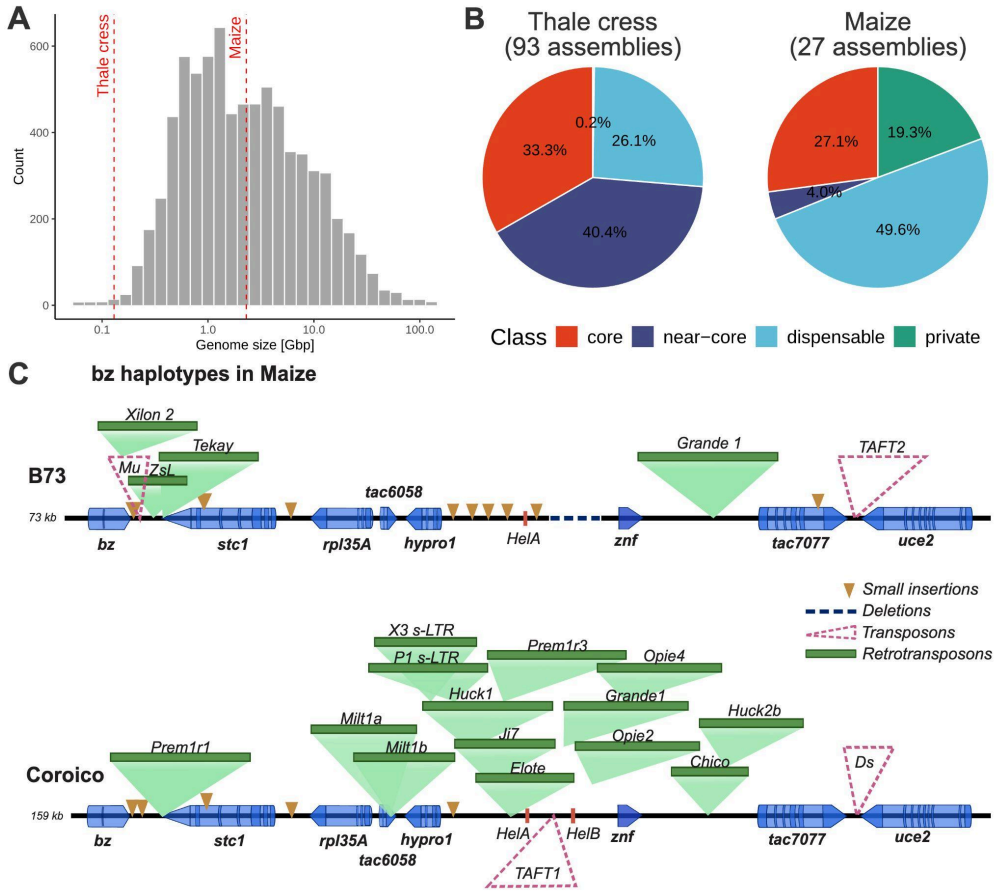


Figure 1. Genome complexity in plants. (A) Distribution of haploid genome sizes (in billions of base pairs); data from the Plant DNA C-values database[64]. (B) Proportion of genes in the pangenomes of *Arabidopsis thaliana* (Thale cress)[65] and *Zea mays* (Maize)[36]. (C) Sequence variation near the bronze (*bz*) gene in two maize lines; adapted from [66]. Blue polygons represent genes. Large triangles point to the insertion sites of DNA (dotted pink) or RNA (green) transposable elements, while dotted black lines and small triangles represent insertions and deletions, respectively.

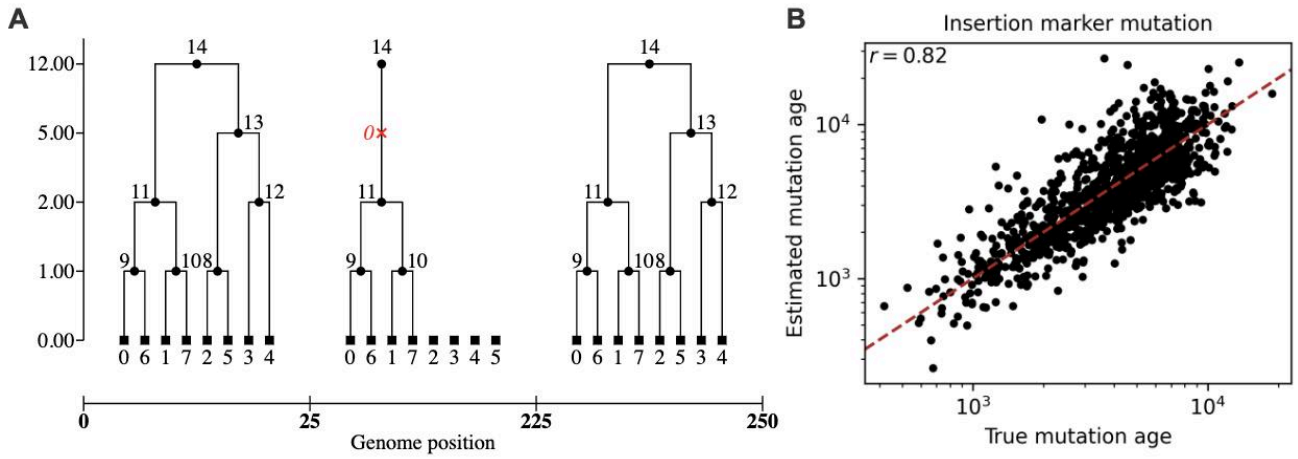


Figure 2. Whole-genome genealogies with presence-absence variation. (A) SLiM[67] simulated genealogy with insertion spanning from position 25 to 225. Tips (squares) represent haplotypes sampled from the population. Some leaves (2,3,4,5) do not have this insertion and thus have no genealogical history within the region. (B) Application of a proof-of-concept of the approach delineated in Aim 1 to estimate insertion age. The scatterplot shows the estimated insertion age using our heuristic approach with SINGER[15] on the y-axis and the true insertion ages on the x-axis for a thousand simulations.

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